The Decomposition Penalty: Quantifying the Impact of Classical Pre-processing on Time-Series Foundation Models in Load Forecasting

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***Abstract*—The integration of Time-Series Foundation Models into short-term load forecasting presents new opportunities for modern smart grid management. However, traditional power grid forecasting heavily relies on Seasonal-Trend (STL) decomposition as a mathematical pre-processing step. This study investigates the empirical impact of classical decomposition on zero-shot foundation models (Chronos-T5) compared to traditional autoregressive gradient boosting (XGBoost). Testing across 107 rolling windows of hourly power load data, we identify a severe "decomposition penalty." Stripping seasonality and trend degraded the foundation model's Mean Absolute Percentage Error (MAPE) from 9.25% to 13.15%. Conversely, a computationally efficient, purely autoregressive XGBoost model achieved a superior 7.68% MAPE. The empirical results suggest that classical decomposition destroys the macro-structural context that foundation models rely on, and that tree-based ensembles currently remain a highly robust, computationally efficient architecture for load forecasting.**

***Index Terms*—Load Forecasting, Time-Series Foundation Models, XGBoost, STL Decomposition, Smart Grids**

I. Introduction

As power grids grow increasingly complex, the volatility of load demands necessitates highly accurate Short-Term Load Forecasting (STLF). Fast, reliable advisory systems are increasingly required at the substation level to execute peak load shaving and prevent grid instability.

Historically, the power sector has relied on statistical decomposition methodologies, such as Seasonal-Trend decomposition using LOESS (STL), to isolate residual signals before applying predictive algorithms. Simultaneously, the artificial intelligence sector is experiencing a paradigm shift toward large Time-Series Foundation Models (TSFMs) like Chronos, which are pre-trained on millions of sequences.

A critical operational question arises when deploying these models: *Does classical decomposition improve foundation model accuracy by isolating the stochastic load, or does it degrade performance by destroying the raw structural context?*

This paper presents an empirical ablation study testing the interaction between STL decomposition and a 46-million parameter foundation model (Chronos-T5-small), benchmarked against computationally lightweight tree-based models (XGBoost) and classical statistical methods (AutoETS).

II. Methodology

A. Data Engine and Evaluation Protocol

The study utilizes the PJM East hourly load dataset. To mitigate concept drift while maintaining sufficient historical context for deep learning layers, the training data is strictly bounded to the years 2015–2017.

Testing is conducted over the year 2018 using a rigorously non-overlapping rolling window evaluation. The forecast horizon is set to 48 hours with a context lookback window of 336 hours (14 days). The evaluation stride is fixed at 48 hours, resulting in 107 discrete testing windows. This strict separation prevents future data leakage and realistically simulates a dynamic grid dispatcher interface.

B. Classical Decomposition Pipeline

The mathematical decomposition pipeline processes the raw load signal (Y_t) into independent vectors before inference:

1) **Base Fit:** STL decomposition ($period = 168$) is applied to the 336h context window.
2) **Trend & Seasonality:** A linear regression projects the trend forward 48h, while the final 168h seasonal block is repeated for the target horizon.
3) **Day of Week (DoW):** To prevent look-ahead bias, historical averages of STL residuals per DoW are calculated strictly on the training set and applied via calendar lookup.
4) **Residual Processing:** The remaining signal is isolated as the target vector for the neural architecture.

C. Model Architecture and Fine-Tuning

The foundation model evaluated is `amazon/chronos-t5-small`. The PyTorch training loop utilizes automatic mixed precision (`torch.amp.autocast`), gradient checkpointing, and gradient accumulation (effective batch size of 16). For the fine-tuning ablation stage, Low-Rank Adaptation (LoRA) is applied (Rank=16, Alpha=32) targeting the query, key, value, and output attention matrices.

The primary machine learning baseline is a purely autoregressive XGBoost regressor, utilizing historical lags ($t -$

$1 \ldots t-24, t-168$) wrapped in a multi-output regressor to predict the 48-hour horizon simultaneously.

III. Results and Ablation Study

Table I details the Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) averaged across the 107 rolling test windows. (Note: All reported performance metrics are truncated to two decimal places for brevity.)

TABLE I
Ablation Study: 107-Window Rolling Forecast Error

Model Configuration	MAE	RMSE	MAPE (%)	Std (%)
Seasonal Naïve	3622.1	4175.4	11.11	7.08
AutoETS	4095.8	4733.1	13.12	5.58
XGBoost (Autoregressive)	**2485.9**	**3035.9**	**7.68**	**4.87**
Raw Chronos (Zero-shot)	3006.2	4032.1	9.25	4.05
STL + Chronos	4148.2	4698.7	12.86	8.96
STL + DoW + Chronos	4248.4	4866.6	13.21	9.01
STL + DoW + Chronos + LoRA	4228.9	4814.0	13.15	9.19

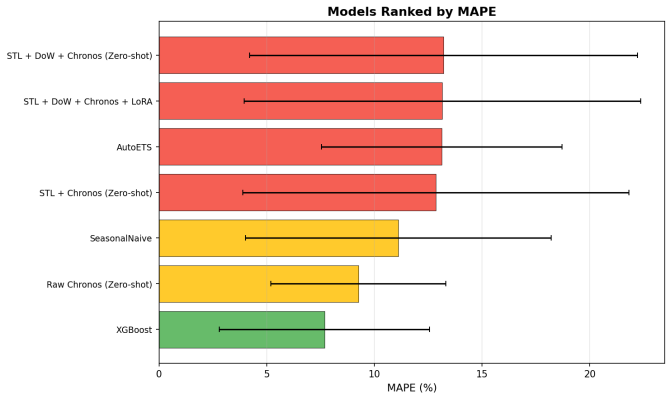

Fig. 1. Model ranking by Mean Absolute Percentage Error (MAPE) across all 107 testing windows, highlighting XGBoost's superiority.

The empirical data demonstrates a severe penalty when coupling classical decomposition with large foundation models. While the raw, zero-shot Chronos model achieved a 9.25% MAPE, mathematically isolating the residuals and filtering calendar variance spiked the error to 13.21%. Furthermore, parameter-efficient fine-tuning (LoRA) across 10 epochs failed to recover the lost signal, yielding a negligible improvement to 13.15%.

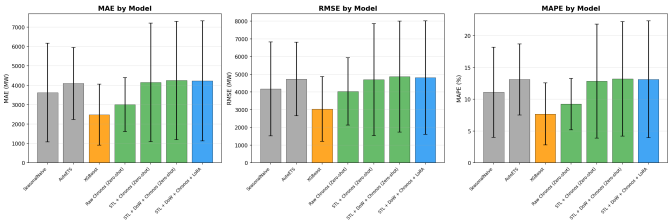

Fig. 2. Ablation study progression demonstrating the degradation of accuracy when applying decomposition techniques to the foundation model.

The traditional XGBoost model drastically outperformed all neural variations, achieving a 7.68% MAPE in a fraction of the compute time.

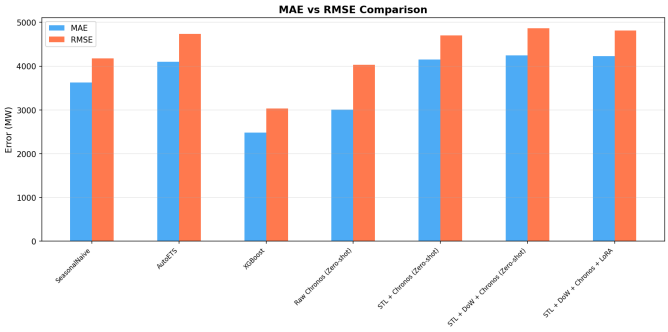

Fig. 3. Comparison of Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) highlighting the absolute deviation metrics across architectures.

IV. Discussion: The Decomposition Penalty

The failure of the STL-Chronos pipeline highlights a fundamental architectural clash between traditional power grid engineering and modern transformer mechanisms.

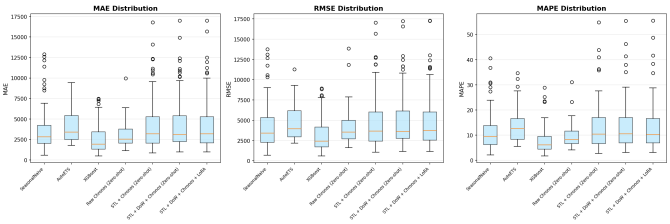

Fig. 4. Error distribution across the 107 rolling windows, highlighting the severe variance instability in the decomposed neural architectures compared to tree-based models.

Time-Series Foundation Models construct their predictive logic using attention layers that scan for macro-shapes: the ramp-up of a morning load, the plateau of an afternoon peak, and the steep drop of an evening trough. By applying STL decomposition, these recognizable features are mathematically flattened. The model is forced to interpret a stationary, highly stochastic residual line devoid of the natural cyclical context it was pre-trained to recognize. We hypothesize that, consequently, the attention heads assign uniform, low-confidence weights to the noise, causing the degradation in accuracy and an explosion in prediction variance (from a baseline Std of 4.05% in the raw model to 9.19% in the decomposed pipeline).

Conversely, tree-based models like XGBoost do not rely on holistic sequence shape recognition. By ingesting strictly autoregressive lags, the decision trees successfully mapped the nonlinear boundaries of the load dynamics, proving highly resilient to the lack of exogenous variables.

V. Conclusion, Limitations, and Future Work

This study establishes that, within the constraints of univariate hourly load forecasting, classical STL decomposition

consistently degrades the performance of Time-Series Foundation Models. For practical smart grid deployments under these parameters, gradient boosted trees remain a highly robust architecture, offering superior accuracy (7.68% MAPE) without the need for complex deep learning infrastructure.

Limitations: We acknowledge several constraints in this preliminary study. First, the empirical evidence is derived from a single univariate dataset (PJM East); verifying whether this decomposition penalty holds across diverse grid topologies and multivariate inputs requires further testing. Second, the parameter-efficient fine-tuning (LoRA) was restricted to 10 epochs without an exhaustive hyperparameter search or convergence analysis, leaving open the possibility that a fully optimized fine-tuning regime could partially recover the decomposed signal. Finally, our hypothesis regarding attention-head weight distribution remains theoretically motivated and requires future mechanistic interpretability studies (e.g., attention rollout visualization) to definitively prove.

Future research will focus on integrating high-frequency multivariate meteorological data to reduce the error threshold, validating these findings across broader datasets, and exploring alternative learned decomposition methods to see if they bypass the penalty observed with classical STL.

VI. Code Availability

To support open science and reproducibility, the complete source code, evaluation datasets, and generated visual artifacts used in this study are publicly available at:
https://github.com/nikhilvinod321/STL-Chronos-Forecasting

VII. Acknowledgment

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